

SYSTEM PROPOSAL: MEDICARE DIRECT

Cloud-Based Clinic & Outpatient Appointment Management Platform

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Target AWS Region:	ap-southeast-1 (Singapore)	VPC Network CIDR:	10.0.0.0/16

1. System Description & Business Objectives (~120 Words)

MediCare Direct is an enterprise outpatient clinic management system designed to digitize patient appointment scheduling, physician availability tracking, and clinical record management. The platform caters to three primary user personas: (1) **Patients** accessing the public web portal to register, search doctor schedules, and reserve appointment slots; (2) **Healthcare Providers and Staff** managing clinical queues, consultation status, and electronic health records (EHR); and (3) **System Administrators** ensuring 99.9% infrastructure availability, automated backup retention, and data security compliance. The major infrastructure requirements include a public-facing web proxy tier, an isolated API logic engine, a KMS-encrypted PostgreSQL database, subnet logical segmentation, stateful firewalls, and role-based IAM controls.

2. System Requirements to AWS Solution Mapping

System Requirement	Proposed AWS Service	Technical Specification & Design Purpose
Public Web Portal	Amazon EC2 (Web Tier)	t3.micro (1 GB RAM, AL2023) running NGINX for public SSL/TLS termination.
Application Processing	Amazon EC2 (App Tier)	t3.small (2 GB RAM, AL2023) executing Node.js Express API logic in private subnet.
Persistent Clinical Data	Amazon EBS (Data Tier)	100 GB gp3 SSD volume encrypted with AWS KMS (AES-256) storing PostgreSQL DB.
Network Isolation	Amazon VPC & Subnets	10.0.0.0/16 VPC divided into Public (10.0.1.0/24), App (10.0.2.0/24), & Data (10.0.3.0/24).
Network Access Control	AWS Security Groups	Stateful firewalls enforcing sg-WebTier -> sg-AppTier -> sg-DBTier port chaining.
Administrative Access	AWS IAM & SSM	Role-based access control with MFA; SSM Session Manager access (SSH Port 22 closed).

3. Architectural Summary & Alignment with Cloud Design Lenses

The proposed architecture strictly implements the 5 AWS Cloud Architectural Lenses: **(1) Segmentation:** Web, application, and database tiers reside in distinct subnets, isolating patient records from external threats. **(2) Least Privilege:** Ingress rules restrict traffic strictly to required ports (80/443 for web, 5000 for app, 5432 for DB). **(3) Controlled Connectivity:** Ingress is bounded by an Internet Gateway, while private subnets have no direct Internet routes. **(4) Scalability:** /24 CIDR blocks accommodate 250+ host addresses per subnet for future Auto Scaling and Load Balancers. **(5) Simplicity:** Omits unneeded complex managed services to keep operational overhead and costs low (\$75.98/month baseline).